

Optimizing Phishing Detection: Comparative Analysis of Lightweight Machine Learning and Transformer Models

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Abstract—Phishing emails continue to be a major cybersecurity concern, prompting the need for solutions that effectively balance precision, efficiency, and expense. This research evaluates several phishing detection models, namely Naive Bayes, Dandelion-optimized Naive Bayes, BERT, and DistilBERT. Although DistilBERT provides impressive accuracy at 99.48%, it demands significant computational power and entails high costs, making it less viable for environments with limited resources. Conversely, Dandelion-optimized Naive Bayes achieves an accuracy of 98.68% with reduced costs and faster processing. For every 10,000 phishing emails, DistilBERT is twice as precise but costs three times more and is four times slower compared to Dandelion. This research underscores the essential role of phishing education, training, and awareness (PETA) programs, which can boost email phishing detection by 20%. Addressing phishing threats, including AI-based methods like deepfakes, effectively requires combining lightweight models with user-focused security measures.

Index Terms—Machine learning, optimization, phishing detection

I. INTRODUCTION

Phishing emails pose a major threat to cybersecurity, requiring tools that adeptly balance speed, precision, and computational load [1]. While lightweight machine learning (ML) models, such as Naive Bayes, offer computational efficiency, they need improvements like the Dandelion algorithm to achieve competitive accuracy, providing an effective solution for organizations with limited resources [2], [3].

On the other hand, transformer models such as Bidirectional Encoder Representations from Transformers (BERT) and its leaner version, DistilBERT, lead the way in phishing detection because they can recognize complex text patterns [4], [5]. Nonetheless, due to substantial computational and hardware requirements, their use is restricted in resource-limited environments [6].

Although machine learning and transformer models play a crucial role in identifying phishing threats, they are insufficient on their own. A holistic defense strategy against phishing must incorporate human elements, such as user behavior and education, highlighting the importance of a comprehensive approach that involves continuous user training [7], [8]. This

research examines the pros and cons of employing ML and resource-demanding transformer models, stressing the importance of synchronizing technological advancements with effective phishing education, training, and awareness (PETA) programs to develop a strong and adaptable cybersecurity system.

The rest of this paper is structured as follows. Section II reviews related work. Section III details our methodology. Section IV presents the results of our comparative analysis, highlighting the effectiveness of each model in detecting phishing emails. Finally, Section V concludes the paper.

II. RELATED WORK

TABLE I
SUMMARY OF RELATED WORK.

Category	Key Differences	References
ML	Focus on specific classification tasks, such as spam detection. Evaluate Naive Bayes within a narrow scope. Do not compare to transformer models or visit real-world phishing applications.	[3], [9]
Transformer Based	Focuses on transformer techniques like embeddings and model refinement. Lacks comparisons to lightweight models or practical trade-offs for resource-constrained use.	[10], [11]
Word cloud	Used for visualizing text patterns. Not used to depict how models like Naive Bayes differentiate phishing from legitimate emails.	[12], [13]
PETA	Studies focus on user behavior and training but isolate this from model evaluations.	[8], [7]

Previous studies on phishing email detection typically categorize into ML, transformer-based models, word clouds, and PETA. These studies often evaluate technologies within these silos without cross-disciplinary comparisons or integration of human factors, which limits their application in real-world settings [3], [9], [10], [11], [12], [13], [8], [7]. A summary of key related studies is given in Table I.

In the realm of ML, research has often focused narrowly on optimizing algorithms like Naive Bayes for specific tasks

such as spam detection, neglecting broader phishing detection implications or the comparison to advanced models like DistilBERT [3], [9], [14]. Similarly, studies on transformer-based models have concentrated on refining techniques for enhanced accuracy in spam filtering without considering their practical deployment in resource-limited environments or their comparison against lightweight ML models [10], [11].

The application of word clouds in research has been limited to visualizing text patterns in non-phishing contexts like product reviews or student feedback. This approach misses opportunities to apply these visualizations in phishing detection [12], [13]. On the human factors front, while there is significant emphasis on the effectiveness of phishing training programs, there is a gap in research concerning how these programs can be effectively integrated/leveraged with technical detection systems to enhance overall defense strategies [8], [7].

This study aims to bridge these gaps by providing a comprehensive analysis that evaluates the effectiveness of both ML and transformer-based models, including their optimized versions, in the context of phishing email detection. It combines technical performance evaluations with practical considerations and human factors, aiming to develop a balanced strategy that integrates technological solutions with user training to combat evolving phishing threats effectively.

III. METHODOLOGY

Algorithm 1 Optimized phishing email detection with ML and transformers.

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1: Input: Dataset  $D$  containing emails with features and labels.
2: Output: Optimally trained model for detecting phishing emails.
3: procedure PREPROCESSDATA( $D$ )
4:   Extract and clean text from  $D$ .
5:   Apply tokenization and remove stopwords.
6:   Convert text to vectors using TF-IDF.
7:   Split  $D$  into training set  $D_{train}$  and testing set  $D_{test}$ .
8: end procedure
9: procedure TRAINMODEL( $D_{train}$ , ModelType)
10:  if ModelType is "NaiveBayes" then
11:    Train Naive Bayes model with Dandelion optimization.
12:  else if ModelType is "BERT" then
13:    Fine-tune BERT model specifically for phishing detection.
14:  else
15:    Fine-tune DistilBERT model for resource efficiency.
16:  end if
17:  return Trained model  $M$ 
18: end procedure
19: procedure EVALUATEMODEL( $M$ ,  $D_{test}$ )
20:  Evaluate  $M$  on  $D_{test}$  to calculate metrics: accuracy, F1-score.
21:  return Evaluation results.
22: end procedure
23: procedure MAIN
24:   $D_{train}, D_{test} \leftarrow$  PREPROCESSDATA( $D$ )
25:   $M_{NB} \leftarrow$  TRAINMODEL( $D_{train}$ , "NaiveBayes")
26:   $M_{BERT} \leftarrow$  TRAINMODEL( $D_{train}$ , "BERT")
27:   $M_{DistilBERT} \leftarrow$  TRAINMODEL( $D_{train}$ , "DistilBERT")
28:  Metrics for  $M_{NB}, M_{BERT}, M_{DistilBERT} \leftarrow$  EVALUATE-
    MODEL( $M_{NB}, M_{DistilBERT}, M_{BERT}, D_{test}$ )
29:  Analyze results and select the model that offers the best performance
    with acceptable resource use.
30: end procedure

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This study evaluates the effectiveness of various phishing detection models, including Naive Bayes, a Dandelion-

optimized Naive Bayes variant, BERT, DistilBERT, and word cloud visualizations as highlighted in Algorithm 1.

A. Data Preprocessing and Experimental Setups

We utilized a balanced dataset from Kaggle, which comprises approximately 82,500 emails categorized as phishing (42,891) and legitimate (39,595). Given its structured and publicly available nature, the Enron dataset was used solely for legitimate (ham) emails. Since phishing emails were scarce during Enron, phishing samples were sourced from more recent collections, including Kaggle and Ling-Spam. This combination ensured a balanced and representative dataset, mitigating concerns about outdated phishing examples. This dataset, enriched with Enron and Ling collections data, provides a robust foundation for evaluating both traditional and advanced ML models [15]. Additionally, we incorporated findings from Marshall et al. [7] on the effectiveness of PETA programs, which helped evaluate how training complements detection methods.

The experimental setup utilized Python in a Jupyter Notebook environment with libraries such as Scikit-learn, TensorFlow, and Hugging Face Transformers [16]. The training was performed on a 12th Gen Intel(R) Core(TM) i7-12700 CPU and an NVIDIA GeForce RTX 3080 Ti 12GB GPU with CUDA acceleration, significantly improving processing efficiency for transformer models like BERT and DistilBERT. The study evaluated lightweight models, including Naive Bayes and its Dandelion-optimized variant, as well as resource-intensive transformer models for phishing email detection.

The model performance evaluation was thorough, employing accuracy, precision, recall, and F1-score metrics to address the unique challenges of phishing detection. The F1 score was particularly emphasized due to its relevance in balancing precision and recall in potentially imbalanced phishing datasets. Cross-validation techniques were employed to ensure the models' reliability and prevent overfitting, with confidence intervals assessed to ensure consistency across different data splits [17], [18].

B. Model Selection and Training

We leveraged both traditional ML and transformer-based models for phishing detection. Naive Bayes and its Dandelion-optimized variant were selected to handle large datasets with minimal computational resources efficiently. In contrast, BERT and DistilBERT were fine-tuned to leverage deep learning for capturing nuanced text patterns in phishing emails.

1) *Naive Bayes*: Naive Bayes assumes that features in a dataset are independent given the class label, simplifying text data into a bag of words for classifying emails. This model efficiently handles high-dimensional spaces with minimal resources, serving as an effective baseline for comparing more complex ML frameworks like transformers [9].

Fine-tuning: The implementation involves preprocessing the email content with a TF-IDF (Term Frequency-Inverse Document Frequency) vectorizer, which quantifies the importance of each word for distinguishing between phishing

and legitimate messages. The vectorizer limits itself to the top 5,000 most significant features, optimizing the balance between performance and computational load. The data is partitioned into training (70%) and testing (30%) sets, comprising 57,740 and 24,746 samples, respectively.

2) *Naive Bayes with Dandelion Optimization*: The Dandelion algorithm, inspired by how dandelion seeds disperse, optimizes the spread of solutions across a problem's search space. It combines exploratory and exploitative strategies to iteratively refine solutions iteratively, adapting effectively to different environments. This balance makes it suitable for complex optimization scenarios [3].

Metaheuristic optimization: As a metaheuristic, the Dandelion algorithm operates by approximating solutions to complex problems through strategies that mimic natural processes, offering a stark contrast to exhaustive optimization methods that attempt to evaluate every possible solution. This strategic navigation of the search space enhances the Naive Bayes model's performance while conserving computational resources [19], [3].

Implementation: The model's implementation involves a simulated dispersal and selection process that effectively improves the classifier's ability to detect phishing emails, optimizing for higher accuracy and robustness without demanding excessive computational power [3].

3) *Word Cloud Analysis*: Word clouds visually map word frequency, aiding in analyzing textual patterns in data [13]. This study generated word clouds for legitimate (ham) and phishing (spam) emails to identify linguistic differences. Ham emails commonly include technical terms like "virus," "submission," and "total," while spam emails feature urgent, promotional words such as "news," "daily," and "top." These visualizations enhance the Naive Bayes model's classification accuracy by confirming its probabilistic calculations to distinguish between ham and spam, validating the model's effectiveness in email categorization [9].



Fig. 1. Word Clouds illustrate differences in language patterns.

Word clouds were generated using the Python WordCloud library to visualize the frequency of words in ham and spam emails. The preprocessing involved categorizing emails into

ham (label 0) and spam (label 1), eliminating irrelevant words, adding custom stopwords, and applying distinct color schemes to differentiate the categories (Fig. 1).

4) *BERT*: The BERT model, developed by Google for natural language processing tasks, employs a bidirectional approach to text analysis. This method enhances its ability to interpret the context of words, making it highly effective for tasks like phishing email detection [20], [4].

Fine-tuning: The dataset was tokenized using the BERT tokenizer into sequences of up to 512 tokens and split into an 80% training and 20% validation split. Training preparation utilized the PyTorch framework, while fine-tuning was conducted using the Hugging Face Transformers library with specific hyperparameters: learning rate of 2×10^{-5} , batch sizes of 16 for training and 64 for evaluation, a maximum of 10 epochs, enabled early stopping, and mixed precision training for enhanced processing efficiency. An NVIDIA GeForce RTX 3080 GPU significantly reduced training time to 48 minutes, a notable improvement over CPU-based methods [21], [16].

5) *DistilBERT: Lighter-Weight Transformer-based Model*: DistilBERT, a distilled version of BERT, is engineered to decrease computational demands while maintaining efficacy through knowledge distillation [5]. It retains about 97% of BERT's linguistic capabilities with 40% fewer parameters and operates 60% faster, making it suitable for resource-sensitive tasks like phishing detection.

Fine-tuning: DistilBERT's compatibility with the dataset was ensured using its tokenizer. The dataset was split into 80% training and 20% validation segments, with fine-tuning done via the Hugging Face Transformers. Key hyperparameters included a learning rate of 3×10^{-5} , batch sizes of 16 for training and 64 for evaluation, 10 epochs, and early stopping and mixed precision training to optimize learning [16], [4]. Training on an NVIDIA GeForce RTX 3080 GPU, DistilBERT, reduced training time by 20% compared to BERT.

6) *Phishing Education, Training, and Awareness (PETA)*: PETA plays a crucial role in addressing sophisticated phishing threats by continuously adapting training to meet the challenges of evolving attack strategies [22]. Effective training programs can bridge the minor performance gaps between detection models, enhancing organizational resilience without relying solely on technological solutions.

Training significance: Comprehensive training is essential for effective phishing defense, as no model can perfectly identify phishing attempts. Organizations must train users to recognize deceptive elements and verify email sources, while regular phishing simulations help build preparedness and awareness. Periodically emphasizing phishing threats maintains vigilance and underlines the importance of careful email examination. Additionally, a reward and penalty system promotes responsible behaviors and accountability.

Training modalities: Various training modalities enhance defenses against phishing: Embedded training integrates realistic simulations into daily tasks to boost detection skills; Gamification uses interactive elements to engage users actively; E-Learning emphasizes critical evaluation to curb impulsive

responses; and Instructor-led training provides hands-on experiences and direct feedback for complex scenarios.

TABLE II
COMPARISON OF ML MODELS FOR PHISHING EMAIL DETECTION

Aspect	Naive Bayes	NB + Dandelion
Type	Probabilistic	Probabilistic +
Feature Handling	Manual	Optimized
Complexity	Low	Moderate
Performance	Basic Tasks	Noisy Data
Training Speed	10 min	10 min
Inference Speed	Very Fast	Very Fast
Interpretability	High	High
Applications	Spam Filtering	Phishing
Resources	CPU	CPU
Notes	Limited	Balanced

TABLE III
COMPARISON OF TRANSFORMER MODELS FOR PHISHING EMAIL DETECTION

Aspect	BERT	DistilBERT
Type	Transformer	Transformer (Reduced)
Feature Handling	Automatic	Automatic
Complexity	High	Moderate
Performance	Complex Text	Complex Text
Training Speed	48 min	40 min
Inference Speed	Moderate	Faster
Interpretability	Low	Low
Applications	NLP Tasks	Speed-Critical NLP
Resources	GPU	GPU
Notes	Intensive	Intensive

C. State-of-the-Art Comparison of Models

Further insights into each model’s specific characteristics, advantages, and limitations are elaborated in Tables II and III. These tables underscore the trade-offs between accuracy, computational demand, and the complexity inherent to each modeling approach, highlighting their respective efficiencies in phishing email classification.

TABLE IV
PERFORMANCE METRICS FOR NAIVE BAYES, DANDELION, BERT, AND DISTILBERT MODELS

Metric	Naive Bayes	Dandelion	BERT	DistilBERT
Accuracy	0.9618	0.9868	0.9948	0.9936
Precision	0.9776	0.9869	0.9926	0.9967
Recall	0.9482	0.9867	0.9974	0.9910
F1 Score	0.9627	0.9873	0.9950	0.9939

IV. PERFORMANCE ANALYSIS AND DISCUSSION

This section presents a comparative analysis of the evaluated phishing detection models, including Naive Bayes, Dandelion-optimized Naive Bayes, BERT, and DistilBERT. The models

were assessed based on accuracy, precision, recall, and F1-score, as summarized in Table IV. Additionally, computational efficiency, training speed, and inference time were analyzed to evaluate the feasibility of each model in real-world applications.

A. Overview of Results and Comparison Factors

Table IV summarizes the classification performance of each model. BERT achieved the highest F1-score (99.50%), followed closely by DistilBERT (99.39%) and Dandelion-optimized Naive Bayes (98.63%). However, these accuracy gains come at a high computational cost. Naive Bayes, while slightly less accurate (96.18% F1-score), remains computationally efficient and interpretable. The performance of the models was evaluated across three key areas: *accuracy trade-offs*, *computational efficiency*, and *inference time*.

B. Comparative Performance Analysis

The evaluation of phishing detection models highlights key performance trade-offs between accuracy, computational cost, and efficiency. Table IV summarizes the classification performance of each model.

Naive Bayes achieved an accuracy of 96.18%, with a precision of 97.76%, a recall of 94.82%, and an F1-score of 96.27%. Cross-validation confirmed its consistency, with a mean accuracy of 96.15% and a 95% confidence interval ranging from 95.92% to 96.39%. These results underscore its efficiency in classifying phishing emails, making it a strong choice for resource-constrained environments.

The Dandelion-optimized Naive Bayes model demonstrated significant improvements, achieving an accuracy of 98.68%, precision of 98.86%, recall of 98.39%, and an F1-score of 98.63%. Cross-validation validated a mean accuracy of 96%, demonstrating its enhanced capability over the standard Naive Bayes model. This optimization allows it to approach the accuracy levels of more complex models like DistilBERT while maintaining low computational overhead [3].

BERT exhibited the highest accuracy among all models, reaching 99.48% accuracy, 99.26% precision, 99.74% recall, and an F1-score of 99.50%. This superior performance underscores BERT’s ability to capture intricate text patterns and effectively distinguish between phishing and legitimate emails [21]. However, its computational demands make deployment challenging in resource-limited settings.

DistilBERT, a more efficient variant of BERT, demonstrated an accuracy of 99.36%, precision of 99.67%, recall of 99.10%, and an F1-score of 99.39%. With a lower computational footprint and an evaluation loss of 0.0273, DistilBERT maintains strong classification performance while improving efficiency over BERT.

In addition to ML-based approaches, PETA enhances phishing detection by integrating adaptive training strategies alongside automated detection models, improving overall detection rates by up to 20% [7], while gamification improves engagement, yielding an 18% boost [8]. E-learning enhances phishing recognition by 13%, though effectiveness depends on learner

engagement [7], whereas instructor-led training achieves a 21% improvement through hands-on experience and direct interaction [8]. These findings underscore the need for a hybrid approach, combining technological defenses with structured PETA interventions to reinforce user awareness and strengthen phishing resilience.

While BERT and DistilBERT deliver superior accuracy, their computational requirements limit real-world deployment in constrained environments. Dandelion-optimized Naive Bayes presents a balanced alternative, offering competitive accuracy with significantly reduced resource demands. PETA-based interventions complement these models by improving user awareness and reducing susceptibility to phishing attacks. These findings highlight the need for a hybrid approach that balances model accuracy, efficiency, and practical implementation constraints.

a) *Accuracy vs. Efficiency: Trade-offs:* BERT and DistilBERT yield the highest detection accuracy but at a significantly higher computational cost. Dandelion-optimized Naive Bayes achieves near-transformer-level accuracy (98.68%) with much lower overhead, making it a competitive choice for resource-limited environments. While less accurate, Naive Bayes remains the fastest model suitable for low-power devices or real-time applications.

b) *Computational Cost::* BERT required approximately 48 minutes to train, whereas DistilBERT reduced training time to 40 minutes due to its lighter architecture. Dandelion Naive Bayes trained in about 10 minutes, offering a viable alternative for environments that cannot afford the computational demands of transformers.

c) *Inference Speed::* Naive Bayes and Dandelion are the fastest, enabling real-time phishing detection with minimal computational power. Without computational resources, BERT and DistilBERT require GPU acceleration, making them less practical for real-time systems.

C. Learning Curve Discussion

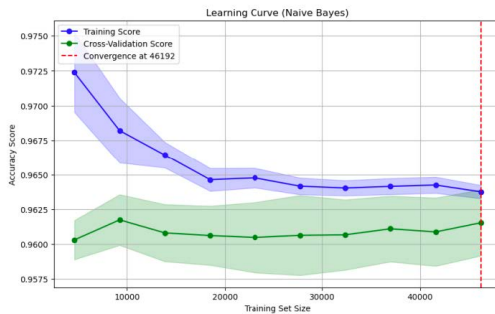


Fig. 2. The training and validation scores converge around 46,192 samples, indicating the sufficiency of training data and the robustness of the model.

Figures 2, 3, 4, and 5 illustrate the learning curves of the evaluated models, highlighting their generalization and convergence behavior.

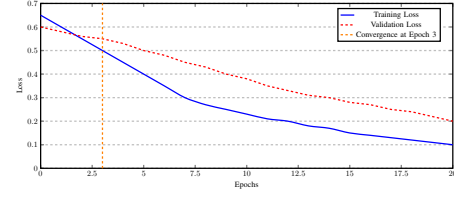


Fig. 3. Learning Curve for Fine-Tuned BERT. The model demonstrates rapid convergence and robust generalization, stabilizing by Epoch 3.

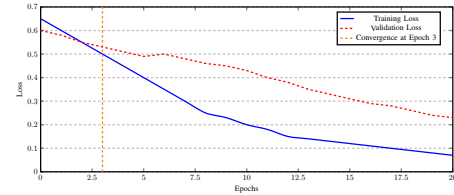


Fig. 4. Learning Curve for Fine-Tuned DistilBERT. The model exhibits rapid convergence, stabilizing by Epoch 3.

Naive Bayes exhibited robust adaptation to increasing dataset sizes, initially achieving a high training score of 97.25%. As more data was introduced, the training score gradually stabilized at 96.2%, demonstrating effective generalization with minimal risk of overfitting [18].

Dandelion-optimized Naive Bayes further improved generalization, converging efficiently with approximately 19,796 training samples. As shown in Fig. 5, additional data beyond this point did not significantly enhance validation accuracy, confirming the optimal training size for maximum effectiveness. The close alignment between training and validation scores underscores the model's robustness and highlights the efficacy of hyperparameter tuning through the Dandelion algorithm [3].

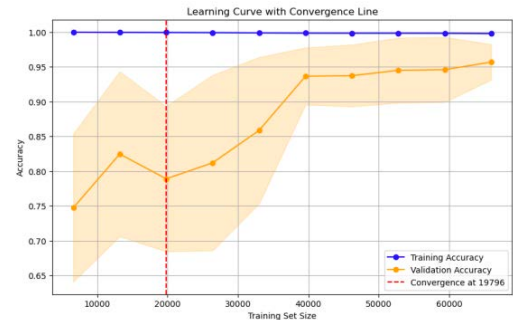


Fig. 5. Learning Curve for Naive Bayes with Dandelion Optimization. The model demonstrates efficient learning and strong generalization.

BERT exhibited a rapid learning process, stabilizing within three epochs as illustrated in Fig. 3. The early stabilization of validation loss and a narrow gap between training and validation losses indicate strong generalization capabilities. However, the high computational cost associated with training and fine-tuning remains a notable constraint [21].

DistilBERT, designed to be a more efficient alternative to BERT, displayed a similarly fast learning rate, reaching convergence by the third epoch (Fig. 4). The consistent decline in training and validation losses suggests efficient pattern learning while mitigating overfitting risks. Given its lower computational demands, DistilBERT offers a favorable trade-off between performance and efficiency for phishing detection [21].

While all models demonstrated effective learning capabilities, their computational efficiency varied significantly. Naive Bayes and its Dandelion-optimized variant provided rapid convergence with minimal computational overhead, making them well-suited for deployment in resource-limited settings. Conversely, BERT and DistilBERT achieved superior accuracy but required substantially greater resources, reinforcing the need for model selection based on the operational constraints of the intended deployment environment.

D. Key Insights and Takeaways

Naive Bayes demonstrated strong phishing detection performance with 96.18% accuracy, offering a computationally efficient solution for resource-limited environments (Table IV). Dandelion-optimized Naive Bayes further improved accuracy to 98.68%, reducing the gap with more complex models while maintaining efficiency [3].

BERT achieved the highest accuracy (99.48%), highlighting its effectiveness in phishing detection, but its computational demands limit its real-world applicability [21]. DistilBERT, while slightly less accurate at 99.36%, provides a more balanced trade-off between accuracy and efficiency.

Visualization techniques, such as word clouds, confirmed key vocabulary distinctions between phishing and legitimate emails, validating the effectiveness of TF-IDF vectorization [9], [13]. Learning curve analysis further demonstrated that Dandelion-optimized Naive Bayes reaches convergence efficiently (Fig. 5), while BERT and DistilBERT stabilize within three epochs but require greater computational power (Fig. 3, Fig. 4).

PETA interventions significantly improved phishing awareness, with embedded training yielding the highest impact (25%) through real-time feedback, reinforcing the necessity of human training alongside automated detection.

PETA improves phishing detection by integrating adaptive training strategies with automated models, increasing detection rates by up to 20% [8]. Instructor-led training offers a 21%

In conclusion, while transformer models provide superior accuracy, Dandelion-optimized Naive Bayes offers a practical alternative balancing performance and efficiency. A hybrid approach incorporating ML models and PETA training ensures a comprehensive phishing defense strategy.

V. CONCLUSION

Phishing attacks are increasingly sophisticated, using AI to exploit human behavior. This study underscores the need to combine technical solutions with robust user training to counter these threats effectively. We see the trade-offs between

cost, efficiency, accuracy, and human factors by comparing ML models from lightweight Dandelion-optimized Naive Bayes to resource-intensive options like BERT and DistilBERT. The results advocate using lightweight models and continuous user education as a scalable, effective strategy to enhance phishing detection and manage resources while adapting to evolving email threats.

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